

Bashing Geometry with Complex Numbers

EVAN CHEN 《陳誼廷》

29 August 2015

This is a (quick) English translation of the complex numbers note I wrote for Taiwan IMO 2014 training. Incidentally I was also working on an airplane.

1 The Complex Plane

Let $\mathbb{C}$ and $\mathbb{R}$ denote the set of complex and real numbers, respectively.

Each $z \in \mathbb{C}$ can be expressed as

$$z = a + bi = r(\cos \theta + i \sin \theta) = re^{i\theta}$$

where $a, b, r, \theta \in \mathbb{R}$ and $0 \leq \theta < 2\pi$. We write $|z| = r = \sqrt{a^2 + b^2}$ and $\arg z = \theta$.

More importantly, each z is associated with a conjugate $\bar{z} = a - bi$. It satisfies the properties

$$\overline{w \pm z} = \bar{w} \pm \bar{z}$$

$$\overline{w \cdot z} = \bar{w} \cdot \bar{z}$$

$$\overline{w/z} = \bar{w}/\bar{z}$$

$$|z|^2 = z \cdot \bar{z}$$

Note that $z \in \mathbb{R} \iff z = \bar{z}$ and $z \in i\mathbb{R} \iff z + \bar{z} = 0$.

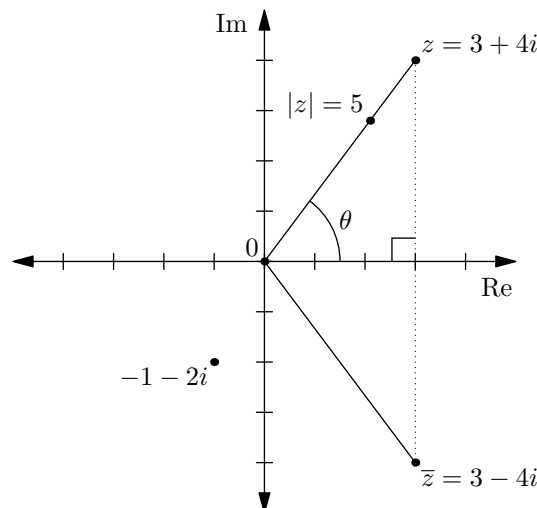


Figure 1: Points $z = 3 + 4i$ and $-1 - 2i$; $\bar{z} = 3 - 4i$ is the conjugate.

We represent every point in the plane by a complex number. In particular, we'll use a capital letter (like Z) to denote the point associated to a complex number (like z).

Complex numbers add in the same way as vectors. The multiplication is more interesting: for each $z_1, z_2 \in \mathbb{C}$ we have

$$|z_1 z_2| = |z_1| |z_2| \text{ and } \arg z_1 z_2 = \arg z_1 + \arg z_2.$$

This multiplication lets us capture a geometric structure. For example, for any points Z and W we can express rotation of Z at W by 90° as

$$z \mapsto i(z - w) + w.$$

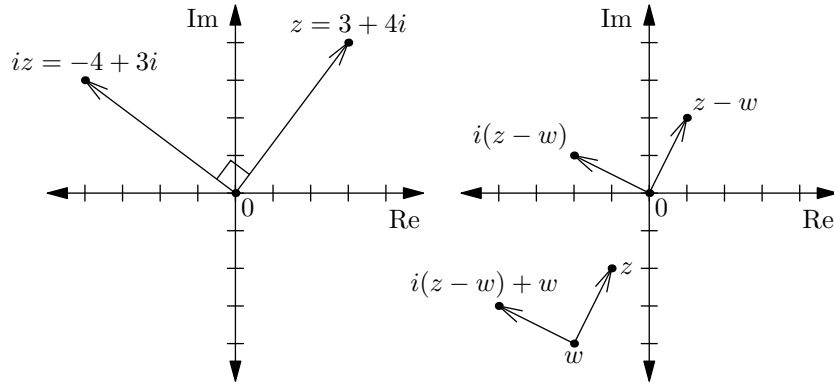


Figure 2: $z \mapsto i(z - w) + w$.

2 Elementary Propositions

First, some fundamental formulas:

Proposition 1. Let A, B, C, D be pairwise distinct points. Then $\overline{AB} \perp \overline{CD}$ if and only if $\frac{d-c}{b-a} \in i\mathbb{R}$; i.e.

$$\frac{d-c}{b-a} + \overline{\left(\frac{d-c}{b-a}\right)} = 0.$$

Proof. It's equivalent to $\frac{d-c}{b-a} \in i\mathbb{R} \iff \arg\left(\frac{d-c}{b-a}\right) \equiv \pm 90^\circ \iff \overline{AB} \perp \overline{CD}$. □

Proposition 2. Let A, B, C be pairwise distinct points. Then A, B, C are collinear if and only if $\frac{c-a}{c-b} \in \mathbb{R}$; i.e.

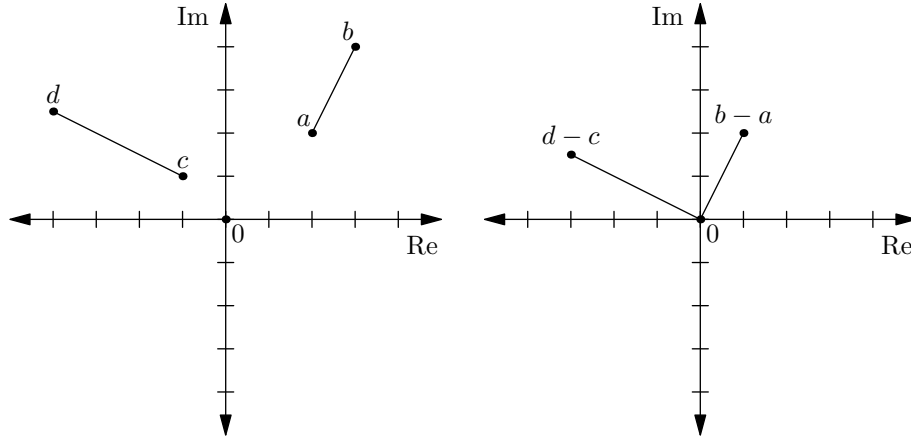
$$\frac{c-a}{c-b} = \overline{\left(\frac{c-a}{c-b}\right)}.$$

Proof. Similar to the previous one. □

Proposition 3. Let A, B, C, D be pairwise distinct points. Then A, B, C, D are concyclic if and only if

$$\frac{c-a}{c-b} : \frac{d-a}{d-b} \in \mathbb{R}.$$

Proof. It's not hard to see that $\arg\left(\frac{c-a}{c-b}\right) = \angle ACB$ and $\arg\left(\frac{d-a}{d-b}\right) = \angle ADB$. (Here angles are directed). □

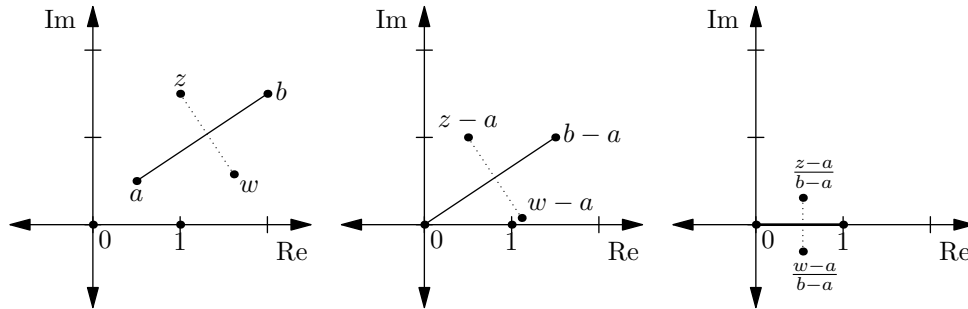
Figure 3: $\overline{AB} \perp \overline{CD} \iff \frac{d-c}{b-a} \in i\mathbb{R}$.

Now, let's state a more commonly used formula.

Lemma 4 (Reflection About a Segment). Let W be the reflection of Z across $\overline{AB}$. Then

$$w = \frac{(a-b)\bar{z} + \bar{a}b - a\bar{b}}{\bar{a} - \bar{b}}.$$

Of course, it then follows that the foot from Z to $\overline{AB}$ is exactly $\frac{1}{2}(w+z)$.

Figure 4: The reflection of Z across $\overline{AB}$.

Proof. According to Figure 4 we obtain

$$\frac{w-a}{b-a} = \overline{\left(\frac{z-a}{b-a}\right)} = \frac{\bar{z}-\bar{a}}{\bar{b}-\bar{a}}.$$

From this we derive $w = \frac{(a-b)\bar{z} + \bar{a}b - a\bar{b}}{\bar{a} - \bar{b}}$. □

Here are two more formulas.

Theorem 5 (Complex Shoelace). Let A, B, C be points. Then $\triangle ABC$ has signed area

$$\frac{i}{4} \begin{vmatrix} a & \bar{a} & 1 \\ b & \bar{b} & 1 \\ c & \bar{c} & 1 \end{vmatrix}.$$

In particular, A, B, C are collinear if and only if this determinant vanishes.

Proof. Cartesian coordinates. □

Often, [Theorem 5](#) is easier to use than [Proposition 2](#).

Actually, we can even write down the formula for an arbitrary intersection of lines.

Proposition 6. Let A, B, C, D be points. Then lines AB and CD intersect at

$$\frac{(\bar{a}b - a\bar{b})(c - d) - (a - b)(\bar{c}d - c\bar{d})}{(\bar{a} - \bar{b})(c - d) - (a - b)(\bar{c} - \bar{d})}.$$

But unless $d = 0$ or a, b, c, d are on the unit circle, this formula is often too messy to use.

3 The Unit Circle, and Triangle Centers

On the complex plane, the **unit circle** is of critical importance. Indeed if $|z| = 1$ we have

$$\bar{z} = \frac{1}{z}.$$

Using the above, we can derive the following lemmas.

Lemma 7. If $|a| = |b| = 1$ and $z \in \mathbb{C}$, then the reflection of Z across $\overline{AB}$ is $a + b - ab\bar{z}$, and the foot from Z to $\overline{AB}$ is

$$\frac{1}{2}(z + a + b - ab\bar{z}).$$

Lemma 8. If A, B, C, D lie on the unit circle then the intersection of $\overline{AB}$ and $\overline{CD}$ is given by

$$\frac{ab(c + d) - cd(a + b)}{ab - cd}.$$

These are much easier to work with than the corresponding formulas in general. We can also obtain the triangle centers immediately:

Theorem 9. Let ABC be a triangle center, and assume that the circumcircle of ABC coincides with the unit circle of the complex plane. Then the circumcenter, centroid, and orthocenter of ABC are given by $0, \frac{1}{3}(a + b + c), a + b + c$, respectively.

Observe that the Euler line follows from this.

Proof. The results for the circumcenter and centroid are immediate. Let $h = a + b + c$. By symmetry it suffices to prove $\overline{AH} \perp \overline{BC}$. We may set

$$z = \frac{h - a}{b - c} = \frac{b + c}{b - c}.$$

Then

$$\bar{z} = \overline{\left(\frac{b + c}{b - c}\right)} = \frac{\bar{b} + \bar{c}}{\bar{b} - \bar{c}} = \frac{\frac{1}{b} + \frac{1}{c}}{\frac{1}{b} - \frac{1}{c}} = \frac{c + b}{c - b} = -z$$

so $z \in i\mathbb{R}$ as desired. □

We can actually even get the formula for the incenter.

Theorem 10. Let triangle ABC have incenter I and circumcircle Γ . Lines AI, BI, CI meet Γ again at D, E, F . If Γ is the unit circle of the complex plane then there exists $x, y, z \in \mathbb{C}$ satisfying

$$a = x^2, b = y^2, c = z^2 \text{ and } d = -yz, e = -zx, f = -xy.$$

Note that $|x| = |y| = |z| = 1$. Moreover, the incenter I is given by $-(xy + yz + zx)$.

Proof. Show that I is the orthocenter of $\triangle DEF$. □

4 Some Other Lemmas

Lemma 11. Let A, B be on the unit circle and select P so that $\overline{PA}, \overline{PB}$ are tangents. Then

$$p = \frac{2}{\bar{a} + \bar{b}} = \frac{2ab}{a + b}.$$

Proof. Let M be the midpoint of $\overline{AB}$ and set $O = 0$. One can show $OM \cdot OP = 1$ and that O, M, P are collinear; the result follows from this. $\square$

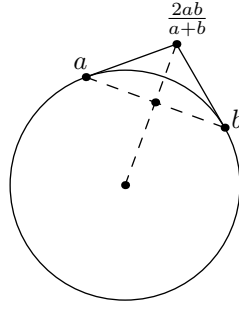


Figure 5: Two tangents. $p = \frac{2}{\bar{a} + \bar{b}}$.

Lemma 12. For any x, y, z , the circumcenter of $\triangle XYZ$ is given by

$$\begin{vmatrix} x & x\bar{x} & 1 \\ y & y\bar{y} & 1 \\ z & z\bar{z} & 1 \end{vmatrix} \div \begin{vmatrix} x & \bar{x} & 1 \\ y & \bar{y} & 1 \\ z & \bar{z} & 1 \end{vmatrix}.$$

This formula is often easier to apply if we shift z to the point 0 first, then shift back afterwards.

5 Examples

Example 13 (MOP 2006). Let H be the orthocenter of triangle ABC . Let D, E, F lie on the circumcircle of ABC such that $\overline{AD} \parallel \overline{BE} \parallel \overline{CF}$. Let S, T, U respectively denote the reflections of D, E, F across $\overline{BC}, \overline{CA}, \overline{AB}$. Prove that points S, T, U, H are concyclic.

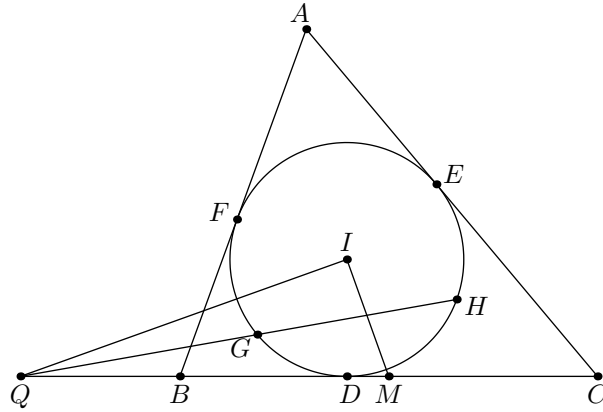
Proof. Let (ABC) be the unit circle and $h = a + b + c$. WLOG, $\overline{AD}, \overline{BE}, \overline{CF}$ are perpendicular to the real axis (rotate appropriately); thus $d = \bar{a}$ and so on. Thus $s = b + c - bc\bar{d} = b + c - abc$ and so on; we now have

$$\frac{s-t}{s-u} = \frac{b-a}{c-a} \quad \text{and} \quad \frac{h-t}{h-u} = \frac{b+abc}{c+abc}.$$

Compute

$$\frac{s-t}{s-u} : \frac{h-t}{h-u} = \frac{(b-a)(c+abc)}{(c-a)(b+abc)} = \frac{\left(\frac{1}{b} - \frac{1}{a}\right) \left(\frac{1}{c} + \frac{1}{abc}\right)}{\left(\frac{1}{c} - \frac{1}{a}\right) \left(\frac{1}{b} + \frac{1}{abc}\right)} \implies \frac{s-t}{s-u} : \frac{h-t}{h-u} \in \mathbb{R}$$

as desired. $\square$



Example 14 (Taiwan TST 2014). In $\triangle ABC$ with incenter I , the incircle is tangent to $\overline{CA}$, $\overline{AB}$ at E , F . The reflections of E , F across I are G , H . Let Q be the intersection of $\overline{GH}$ and $\overline{BC}$, and let M be the midpoint of $\overline{BC}$. Prove that $\overline{IQ}$ and $\overline{IM}$ are perpendicular.

Solution. Let D be the foot from I to $\overline{BC}$, and set (DEF) as the unit circle. (This lets us exploit the results of [Section 3](#).) Thus $|d| = |e| = |f| = 1$, and moreover $g = -e$, $h = -f$. Let $x = \overline{d} = \frac{1}{d}$ and define y, z similarly. Then

$$b = \frac{2}{\overline{d} + \overline{f}} = \frac{2}{x + z}.$$

Similarly, $c = \frac{2}{x+y}$, so

$$m = \frac{1}{2}(b + c) = \frac{1}{x + y} + \frac{1}{x + z} = \frac{2x + y + z}{(x + y)(x + z)}.$$

Next, we have $Q = \overline{DD} \cap \overline{GH}$, which implies

$$q = \frac{dd(g + h) - gh(d + d)}{d^2 - gh} = \frac{\frac{1}{x^2} \left(-\frac{1}{y} - \frac{1}{z} \right) - \frac{1}{yz} \frac{2}{x}}{\frac{1}{x^2} - \frac{1}{yz}} = \frac{2x + y + z}{x^2 - yz}.$$

so

$$m/q = \frac{x^2 - yz}{(x + y)(x + z)}.$$

Now,

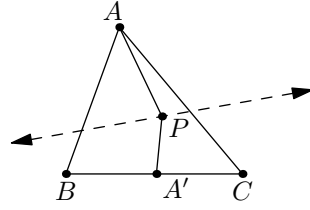
$$\overline{m/q} = \frac{\frac{1}{x^2} - \frac{1}{yz}}{\left(\frac{1}{x} + \frac{1}{y} \right) \left(\frac{1}{x} + \frac{1}{z} \right)} = \frac{yz - x^2}{(x + y)(x + z)} = -m/q$$

thus $m/q \in i\mathbb{R}$, as desired. $\square$

Example 15 (USAMO 2012). Let P be a point in the plane of $\triangle ABC$, and γ a line through P . Let A', B', C' be the points where the reflections of lines PA, PB, PC with respect to γ intersect lines BC, AC, AB respectively. Prove that A', B', C' are collinear.

Solution. Let $p = 0$ and set γ as the real line. Then A' is the intersection of bc and $p\bar{a}$. So, using [Proposition 6](#) we get

$$a' = \frac{\bar{a}(\bar{b}c - b\bar{c})}{(\bar{b} - \bar{c})\bar{a} - (b - c)a}.$$



Note that

$$\bar{a}' = \frac{a(b\bar{c} - \bar{b}c)}{(b-c)a - (\bar{b} - \bar{c})\bar{a}}.$$

Thus by [Theorem 5](#), it suffices to prove

$$0 = \begin{vmatrix} \frac{\bar{a}(b\bar{c} - \bar{b}c)}{(b-c)a - (\bar{b} - \bar{c})\bar{a}} & \frac{a(b\bar{c} - \bar{b}c)}{(b-c)a - (\bar{b} - \bar{c})\bar{a}} & 1 \\ \frac{\bar{b}(c\bar{a} - c\bar{a})}{(\bar{c} - \bar{a})\bar{b} - (c-a)b} & \frac{b(c\bar{a} - c\bar{a})}{(c-a)b - (\bar{c} - \bar{a})\bar{b}} & 1 \\ \frac{\bar{c}(a\bar{b} - a\bar{b})}{(\bar{a} - \bar{b})\bar{c} - (a-b)c} & \frac{c(a\bar{b} - a\bar{b})}{(a-b)c - (\bar{a} - \bar{b})\bar{c}} & 1 \end{vmatrix}.$$

This is equivalent to

$$0 = \begin{vmatrix} \bar{a}(\bar{b}c - b\bar{c}) & a(\bar{b}c - b\bar{c}) & (\bar{b} - \bar{c})\bar{a} - (b-c)a \\ \bar{b}(\bar{c}a - c\bar{a}) & b(\bar{c}a - c\bar{a}) & (\bar{c} - \bar{a})\bar{b} - (c-a)b \\ \bar{c}(\bar{a}b - a\bar{b}) & c(\bar{a}b - a\bar{b}) & (\bar{a} - \bar{b})\bar{c} - (a-b)c \end{vmatrix}.$$

Evaluating the determinant gives

$$\sum_{\text{cyc}} ((\bar{b} - \bar{c})\bar{a} - (b-c)a) \cdot \begin{vmatrix} b & \bar{b} \\ c & \bar{c} \end{vmatrix} \cdot (\bar{c}a - c\bar{a})(\bar{a}b - a\bar{b})$$

or, noting the determinant is $b\bar{c} - \bar{b}c$ and factoring it out,

$$(\bar{b}c - c\bar{b})(\bar{c}a - c\bar{a})(\bar{a}b - a\bar{b}) \sum_{\text{cyc}} (ab - ac + \bar{c}\bar{a} - \bar{b}\bar{a}) = 0. \quad \square$$

Example 16 (Taiwan TST Quiz 2014). Let I and O be the incenter and circumcenter of ABC . A line ℓ is drawn parallel to $\overline{BC}$ and tangent to the incircle of ABC . Let X, Y be on ℓ so that I, O, X are collinear and $\angle XIY = 90^\circ$. Show that A, X, O, Y are concyclic.

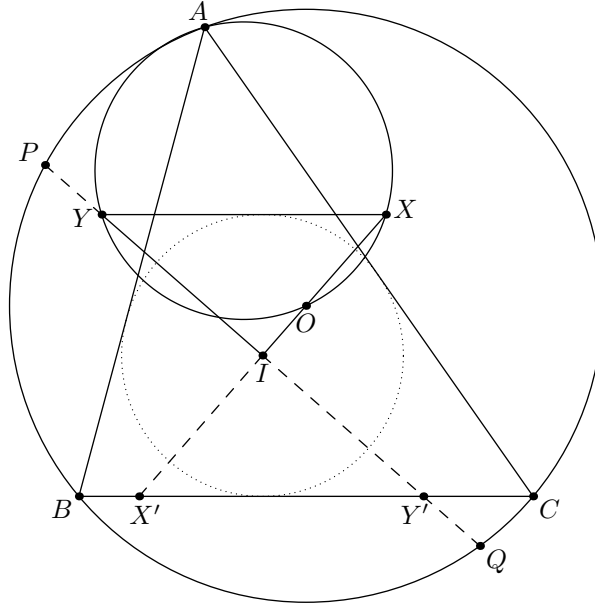
Solution. Let X' and Y' respectively denote the reflections of X and Y across I . Note that X, Y lie on $\overline{BC}$. Also, let P, Q be the intersections of $\overline{IY}$ with the circumcircle.

Of course, (ABC) is the unit circle. Let j be the complex number corresponding to I (to avoid confusion with $i = \sqrt{-1}$). Thus,

$$x' = \frac{(\bar{b}c - b\bar{c})(j - 0) - (\bar{j}0 - j\bar{0})(b - c)}{(\bar{b} - \bar{c})(j - 0) - (b - c)(\bar{j} - \bar{0})} = \frac{j \cdot \frac{c^2 - b^2}{bc}}{j \cdot \frac{c - b}{bc} - (b - c)\bar{j}} = \frac{j(b + c)}{j + bc\bar{j}}.$$

We seek y' now. Consider the quadratic equation in z given by

$$\frac{z - j}{j} + \frac{\frac{1}{z} - \bar{j}}{\bar{j}} = 0 \iff z^2 - 2jz + j/\bar{j} = 0.$$



Its zeros in z are p and q , which implies that $p + q = 2j$ and $pq = j/\bar{j}$ (by Vieta!). From this we can compute

$$y' = \frac{pq(b+c) - bc(p+q)}{pq - bc} = \frac{j(b+c) - 2bcj\bar{j}}{j - bc\bar{j}} = \frac{j(b+c) - 2bcj\bar{j}}{j - bc\bar{j}}.$$

which gives

$$x = 2j - x' = \frac{j(2j - b - c + 2bc\bar{j})}{j + bc\bar{j}} \quad \text{and} \quad y = 2j - y' = \frac{j(2j - b - c)}{j - bc\bar{j}}.$$

From this we can obtain

$$\begin{aligned} y - x &= j \cdot \frac{(2j - b - c)(j + bc\bar{j}) - (2j - b - c + 2bc\bar{j})(j - bc\bar{j})}{(j - bc\bar{j})(j + bc\bar{j})} \\ &= j \cdot \frac{2bc\bar{j}(2j - b - c) - 2bc\bar{j}(j - bc\bar{j})}{(j - bc\bar{j})(j + bc\bar{j})} \\ &= j \cdot \frac{2bc\bar{j}(j - b - c + bc\bar{j})}{(j - bc\bar{j})(j + bc\bar{j})} \\ X &= \frac{y - x}{x} = \frac{2bc\bar{j}(j - b - c + bc\bar{j})}{(j - bc\bar{j})(2j - b - c + 2bc\bar{j})} \\ A &= \frac{y - a}{a} = \frac{j(2j - b - c - a) + abc\bar{j}}{a(j - bc\bar{j})} \end{aligned}$$

We need to prove $X/A = \overline{X/A}$. Now set $a = x^2$, $b = y^2$, $c = z^2$, $j = -(xy + yz + zx)$, $\bar{j} = -\frac{x+y+z}{xyz}$ (this is a different x, y than the points X and Y .) So, the above rewrites as

$$\begin{aligned} X &= \frac{2\frac{yz}{x}(x+y+z)(\frac{yz}{x}(x+y+z) + y^2 + z^2 + xy + yz + zx)}{(-\frac{yz}{x}(x+y+z) + xy + yz + zx)(y^2 + z^2 + 2(xy + yz + zx) + 2\frac{yz}{x}(x+y+z))} \\ &= \frac{2yz(x+y+z)(2xyz + \sum_{\text{sym}} x^2y)}{(y+z)(x^2 - yz)(x(y+z)(2x+y+z) + 2yz(x+y+z))} \end{aligned}$$

$$= \frac{2yz(x+y+z)(x+y)(x+z)}{(x^2-yz)((x^2+yz)(y+z) + (xy+yz+zx)(x+y+z))}$$

and

$$A = \frac{(xy+yz+zx)(x+y+z)^2 - xyz(x+y+z)}{x^2(-(xy+yz+zx) + \frac{yz}{x}(x+y+z))} = \frac{(x+y+z)(x+y)(y+z)(z+x)}{x(yz-x^2)(y+z)}$$

thus

$$\begin{aligned} X/A &= \frac{-2xyz}{(x^2+yz)(y+z) + (x+y+z)(xy+yz+zx)} \\ &= \frac{-\frac{2}{xyz}}{(\frac{1}{x^2} + \frac{1}{yz})(\frac{1}{y} + \frac{1}{z}) + (\frac{1}{x} + \frac{1}{y} + \frac{1}{z})(\frac{1}{xy} + \frac{1}{yz} + \frac{1}{zx})} = \overline{X/A}. \end{aligned} \quad \square$$

6 Practice Problems

1. Let $ABCD$ be cyclic. Let H_A, H_B, H_C, H_D denote the orthocenters of BCD, CDA, DAB, ABC . Show that $\overline{AH_A}, \overline{BH_B}, \overline{CH_C}, \overline{DH_D}$ are concurrent.
2. (China TST 2011) Let Γ be the circumcircle of a triangle ABC . Assume $\overline{AA'}, \overline{BB'}, \overline{CC'}$ are diameters of Γ . Let P be a point inside ABC and let D, E, F be the feet from P to $\overline{BC}, \overline{CA}, \overline{AB}$. Let X be the reflection of A' across D ; define Y and Z similarly. Prove that $\triangle XYZ \sim \triangle ABC$.
3. In circumscribed quadrilateral $ABCD$ with incircle ω , Prove that the midpoint of $\overline{AC}$ and the midpoint of $\overline{BD}$ are collinear with the center of ω .
4. (Simson Line) Let ABC be a triangle and P a point on its circumcircle.
 - (a) Let D, E, F be the feet from P to $\overline{BC}, \overline{CA}, \overline{AB}$. Show that D, E, F are collinear.
 - (b) Moreover, prove that the line through these points bisects $\overline{PH}$, where H is the orthocenter of ABC .
5. (PUMaC Finals) Let γ and I be the incircle and incenter of triangle ABC . Let D, E, F be the tangency points of γ to $\overline{BC}, \overline{CA}, \overline{AB}$ and let D' be the reflection of D about I . Assume EF intersects the tangents to γ at D and D' at points P and Q . Show that $\angle DAD' + \angle PIQ = 180^\circ$.
6. (Schiffler Point) Let triangle ABC have incenter I . Prove that the Euler lines of $\triangle AIB, \triangle BIC, \triangle CIA, \triangle ABC$ are concurrent.
7. (USA TST 2014) Let $ABCD$ be a cyclic quadrilateral and let E, F, G, H be the midpoints of $\overline{AB}, \overline{BC}, \overline{CD}, \overline{DA}$. Call W, X, Y, Z the orthocenters of AHE, BEF, CFG, DGH . Prove that $ABCD$ and $WXYZ$ have the same area.
8. (Iran 2004) Let O be the circumcenter of ABC . A line ℓ through O cuts $\overline{AB}$ and $\overline{AC}$ at points X and Y . Let M and N be the midpoints of $\overline{BY}, \overline{CX}$. Show that $\angle MON = \angle BAC$.
9. (APMO 2010) Let ABC be an acute triangle, where $AB > BC$ and $AC > BC$. Denote by O and H the circumcenter and orthocenter. The circumcircle of AHC intersects AB again at M ; the circumcircle of AHB intersects AC again at N . Prove that the circumcenter of triangle MNH lies on line OH .

10. (Iran 2013) Let ABC be acute, and M the midpoint of minor arc $\widehat{BC}$. Let N be on the circumcircle of ABC such that $\overline{AN} \perp \overline{BC}$, and let K, L lie on AB, AC so that $\overline{OK} \parallel \overline{MB}$, $\overline{OL} \parallel \overline{MC}$. (Here O is the circumcenter of ABC). Prove that $NK = NL$.
11. (MOP 2006) Cyclic quadrilateral $ABCD$ has circumcenter O . Let P be a point in the plane and let O_1, O_2, O_3, O_4 be the circumcenters of PAB, PBC, PCD, PDA . Show that the midpoints of $\overline{O_1O_3}, \overline{O_2O_4}, \overline{OP}$ are concurrent.